SHIVARAMAN T

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Professional Summary

Final-year B.Tech student in Electronics and Communication Engineering, specializing in Artificial Intelligence and Cybernetics, with an **8.68/10 CGPA**. Skilled in Python, SQL, machine learning, data preprocessing, statistical analysis, Linux fundamentals, and application development. Experienced in API integration, software testing, debugging, and cloud-based application deployment. Interested in technology consulting, hybrid cloud, AI/ML, data engineering, Generative AI, and developing scalable technology solutions for business problems.

Education

Vellore Institute of Technology Sep 2023 – May 2027
B.Tech in Electronics and Communication Engineering **CGPA: 8.68/10**
Specialization: Artificial Intelligence and Cybernetics

Technical Skills

Programming and Scripting: Python, SQL, JavaScript, Java, Embedded C
Cloud and Infrastructure: AWS EC2, AWS S3, Linux Fundamentals, Networking Fundamentals, Cloud Deployment, Git, GitHub
AI, Data and Analytics: Machine Learning, Scikit-learn, NumPy, Pandas, SciPy, NLP, spaCy, Data Cleaning, Data Preprocessing, Feature Engineering, Statistical Analysis, Model Evaluation
Databases and Data Handling: Relational Data Structures, SQL, Database Fundamentals, JSON, Structured Data Processing
Application Development and Testing: Flask, React.js, REST APIs, HTML, CSS, API Integration, Postman, Unit Testing, Debugging, Software Validation
Generative AI: LLM Concepts, Generative AI Fundamentals, Prompt Engineering Fundamentals
Professional Skills: Analytical Thinking, Problem Solving, Root Cause Analysis, Technical Documentation, Collaboration, Communication

Experience

MPOnline Limited — *Software Engineering Intern* May 2026 – Jul 2026

- Profiled Python routines and resolved processing bottlenecks, improving component processing speed by **15%** in staging environments.
- Investigated database failures through structured debugging and root cause analysis, reducing runtime exceptions by **25%** through validation.
- Created technical documentation and unit tests, contributing to reported **85% baseline code coverage** and systematic software validation.

Projects

Voice Command Shopping Assistant (VoiceCart) GitHub | Live Demo

- Developed a multilingual web application using JavaScript and Web Speech APIs to process voice commands in English, Hindi, and Tamil.
- Implemented regular-expression-based intent classification, achieving **92% intent execution accuracy** during project testing.
- Implemented browser-side state management and command handling to support responsive interactions.

ECG Signal Analytics and Heart Rate Estimation GitHub

- Processed ECG datasets using Python and scientific computing libraries, applying data cleaning and preprocessing techniques for signal analysis.
- Configured a fourth-order Butterworth filter to attenuate high-frequency noise before signal peak detection and heart-rate estimation.
- Analyzed filtered signal characteristics to derive interpretable heart-rate measurements and compare signals.

Task Management Platform GitHub

- Developed a web application featuring authentication, task assignment, workflow management, and status tracking.
- Designed relational data structures for organized task storage, retrieval, and status management.
- Debugged application errors and investigated root causes to improve task workflow reliability.

Publication

- Co-authored "Smart Helmet Systems: IoT Enhanced Safety and Integrated Display for Two Wheelers," published and presented at ICRAME 2025, NIT Silchar.

Achievements

- **TCS CodeVita Season 13:** Global Rank 1347.
- **AWS Machine Learning Challenge:** Top 100 among 5,000+ participants.
- **Bharatiya Antariksh Hackathon 2024:** Finalist.

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services
Applied Machine Learning in Python – University of Michigan, Coursera